EXPERIMENT NO: 2

(ii) To generate and visualize basic discrete-time signals and analyze their amplitude and time-index characteristics.

Expected Outcomes

- Generate correctly indexed discrete-time sequences.
- Plot sequences using stem representation.
- Identify discrete-time waveform characteristics and periodicity.
- Validate the generated samples against theoretical expressions.

Software / Laboratory Environment

MATLAB

Formulae / Analysis

$x[n] = A \sin(2\pi f_0 n + \phi)$.
Periodic DT signal: $x[n + N_0] = x[n]$ for a positive integer N_0.
$x[n] = a^n u[n]$ for a suitable discrete exponential sequence.
Even sequence: $x[n] = x[-n]$; odd sequence: $x[n] = -x[-n]$.

Experimental Procedure

1. Define the integer sample index n.
2. Generate unit impulse, unit step, ramp, exponential and sinusoidal sequences.
3. Display each sequence using a stem plot.
4. Determine sample amplitudes and period where applicable.
5. Classify the sequence as periodic/non-periodic and even/odd wherever applicable.
6. Compare the generated samples with theoretical values and record observations.

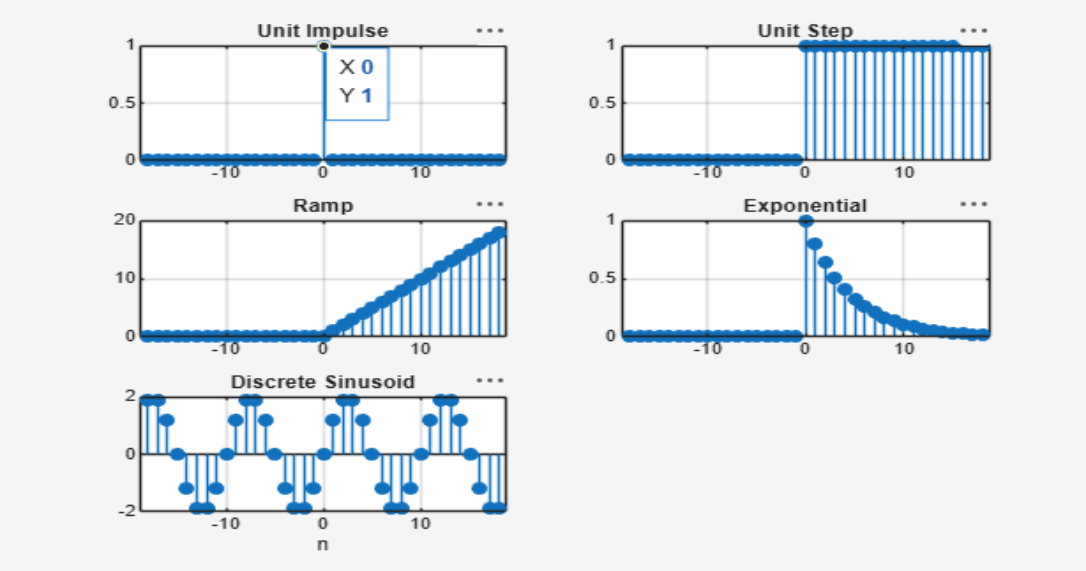
Key Conditions & Threshold Values for Analysis

- Discrete-time signals are defined only at integer indices.
- Stem plots should be used to emphasize discrete samples.
- DT sinusoidal periodicity depends on the normalized frequency satisfying the rationality condition.
- Use sufficient sample range to observe the sequence behavior.

MATLAB Coding

```
% Experiment 2: Generation of Discrete-Time Signals
clc; clear; close all;
n = -20:20;
A = 2; f0 = 0.1; a = 0.1;
imp = double(n==0);
step = double(n>=0);
ramp = n.*step;
expSig = (0.8).^n .* step;
sinSig = A*sin(2*pi*f0*n);
figure;
subplot(3,2,1); stem(n,imp,'filled'); grid on; title('Unit Impulse');
subplot(3,2,2); stem(n,step,'filled'); grid on; title('Unit Step');
subplot(3,2,3); stem(n,ramp,'filled'); grid on; title('Ramp');
subplot(3,2,4); stem(n,expSig,'filled'); grid on; title('Exponential');
subplot(3,2,5); stem(n,sinSig,'filled'); grid on; title('Discrete Sinusoid');
xlabel('n');
```

Input and Output Graphs



Resut

The continuous-time and discrete-time signals were generated and analyzed successfully. The signals were classified as periodic or aperiodic, and the even and odd components were obtained and used to reconstruct the original signal successfully.

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks(10)	Marks obtained
1. Generate and visualize basic continuous-time signals such as unit step, unit impulse, ramp, sinusoidal and exponential.	**CT Signal Generation & MATLAB Implementation**	Generates all five CT signals accurately with a suitable time vector and step size. Uses correct MATLAB commands . Plots are correctly scaled, with titles, axis labels and grid.	Generates the required CT signals and plots correctly, with minor errors in code, time range, parameters or labelling.	Generates basic CT signals but has noticeable errors in code, step or impulse representation, or plots, and needs assistance.	Unable to generate the required CT signals or produce meaningful plots.	**25**	**2.5**	
2. Generate and visualize basic discrete-time signals such as unit sample, unit step, ramp, sinusoidal and exponential sequences.	**DT Signal Generation & MATLAB Implementation**	Generates all five DT sequences accurately with a proper index vector n. Uses stem correctly. Plots are correctly scaled, with titles, axis labels and grid.	Generates the required DT sequences and plots correctly, with minor errors in index range, stem usage, parameters or labelling.	Generates basic DT sequences but has noticeable errors in code, indexing or stem plots, and needs assistance.	Unable to generate the required DT sequences or produce meaningful stem plots.	**25**	**2.5**	
3. Classify CT and DT signals as periodic or aperiodic and identify their waveform characteristics.	**Signal Classification & Periodicity Analysis**	Correctly classifies all signals using DT. Determines the period where applicable. Identifies frequency, phase, and	Classifies most signals correctly and finds most periods, with minor errors or limited justification.	Gives basic classifications . Period calculation or justification is incomplete and needs guidance.	Unable to classify the signals correctly or justify the classification .	**25**	**2.5**	

		growth or decay. Explains the reasoning clearly with mathematical conditions and waveform evidence.						
4. Analyze the amplitude and time-index characteristics of discrete-time signals.	**Amplitude & Time-Index Analysis**	Correctly determines maximum and minimum amplitude, the index n at which each occurs, and signal energy. Correctly shows the effect of time shifting and folding on the index. Interprets the results clearly.	Analyzes most amplitude and index characteristics correctly, with minor errors in values or interpretation.	Gives basic amplitude and index analysis with noticeable errors, and needs guidance.	Unable to analyze the amplitude and time-index characteristics or interpret the results.	**25**	**2.5**	
					Total	**100**	**10**	